CS & IT ENGINEERING

DIGITAL LOGIC

Combinational Circuit



Lecture No. 04



By- CHANDAN SIR

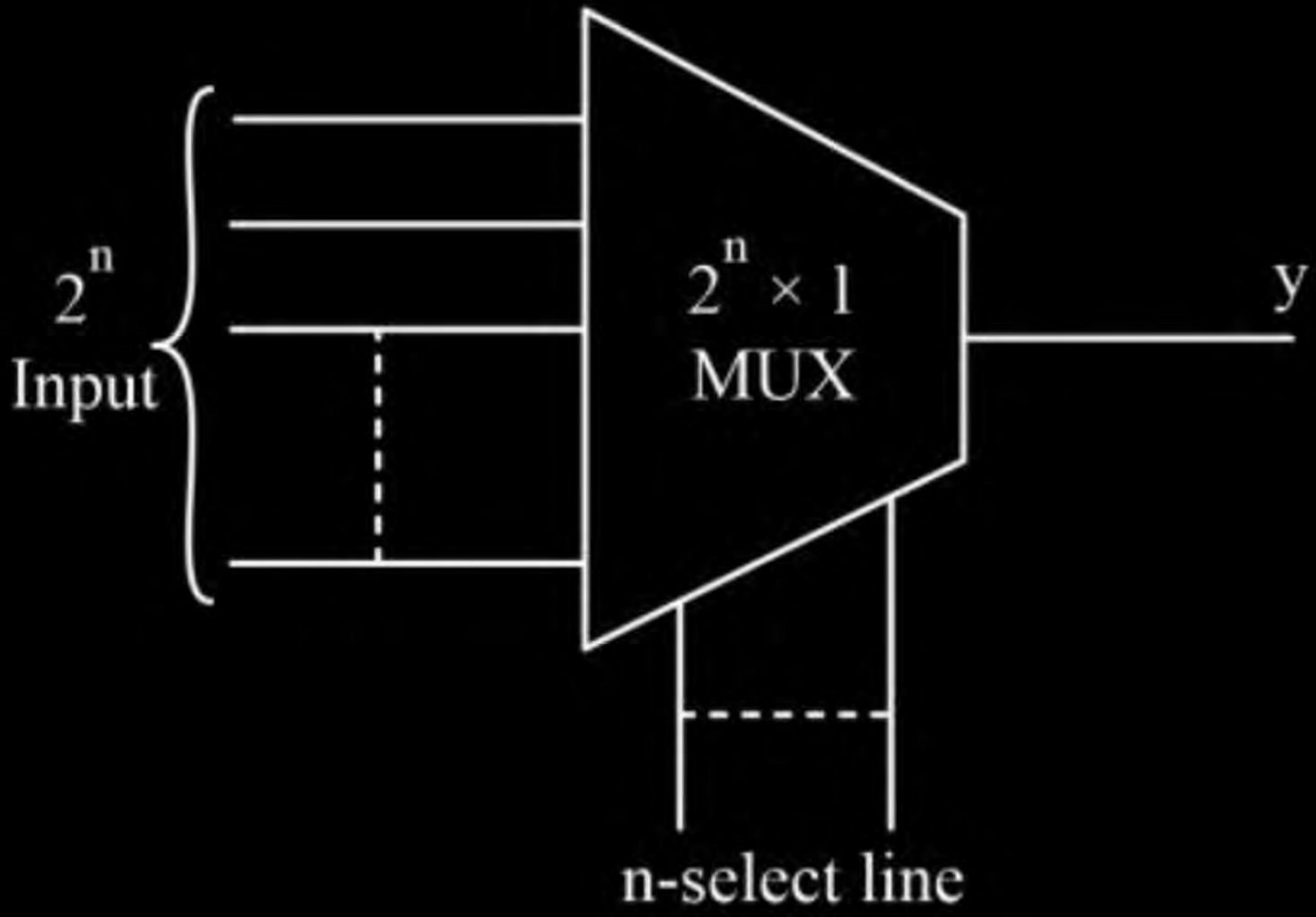


01 MULTIPLEXER

02 QUESTION PRACTICE

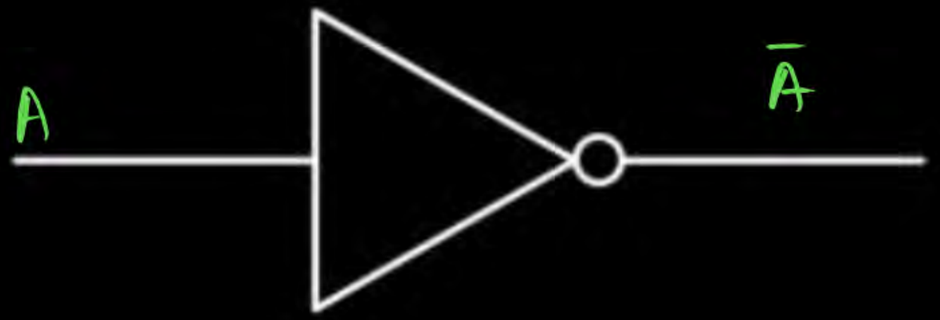
03 DISCUSSION

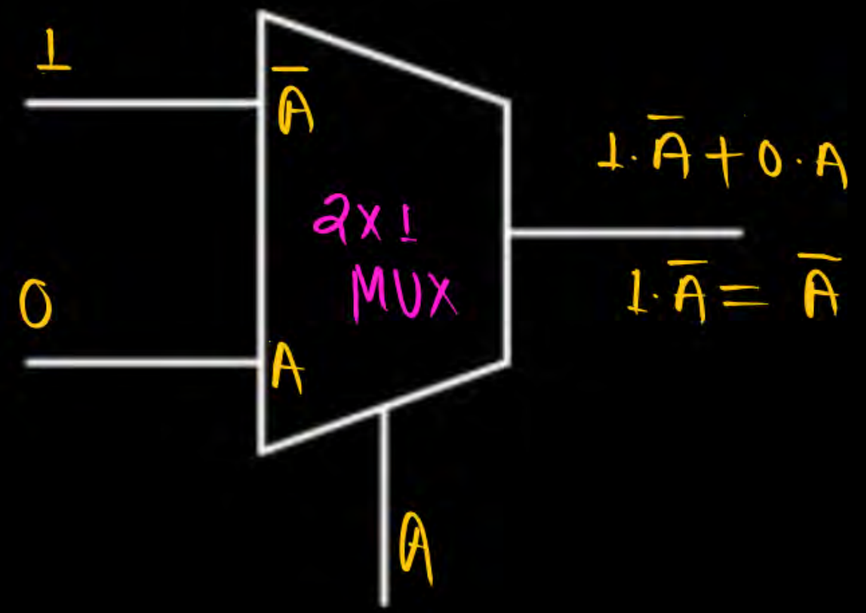
MULTIPLEXER



Type-2 MUX as a Universal Logic

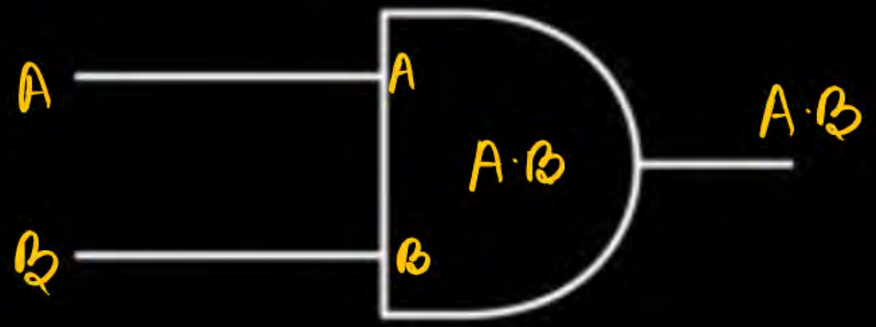
1. Not GATE

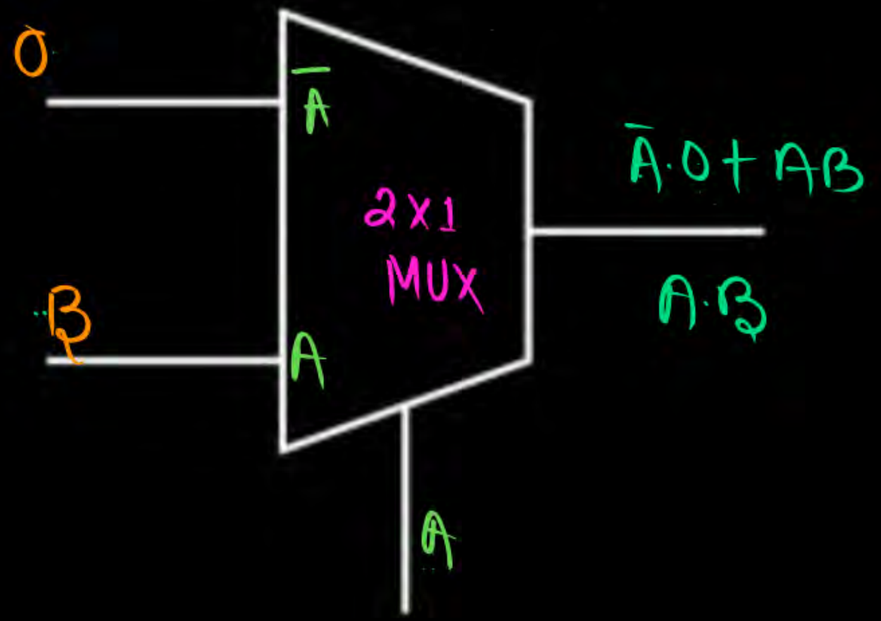




Type-2 MUX as a Universal Logic

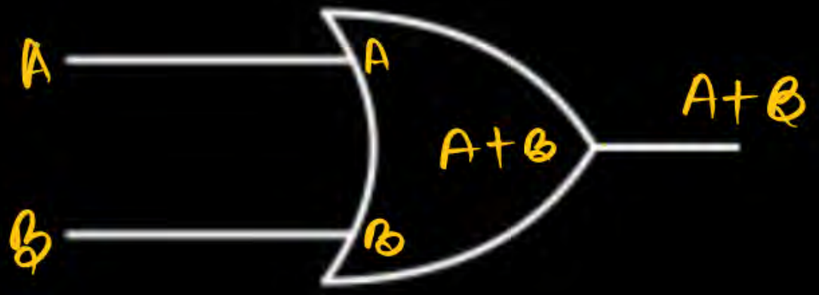
2. AND GATE

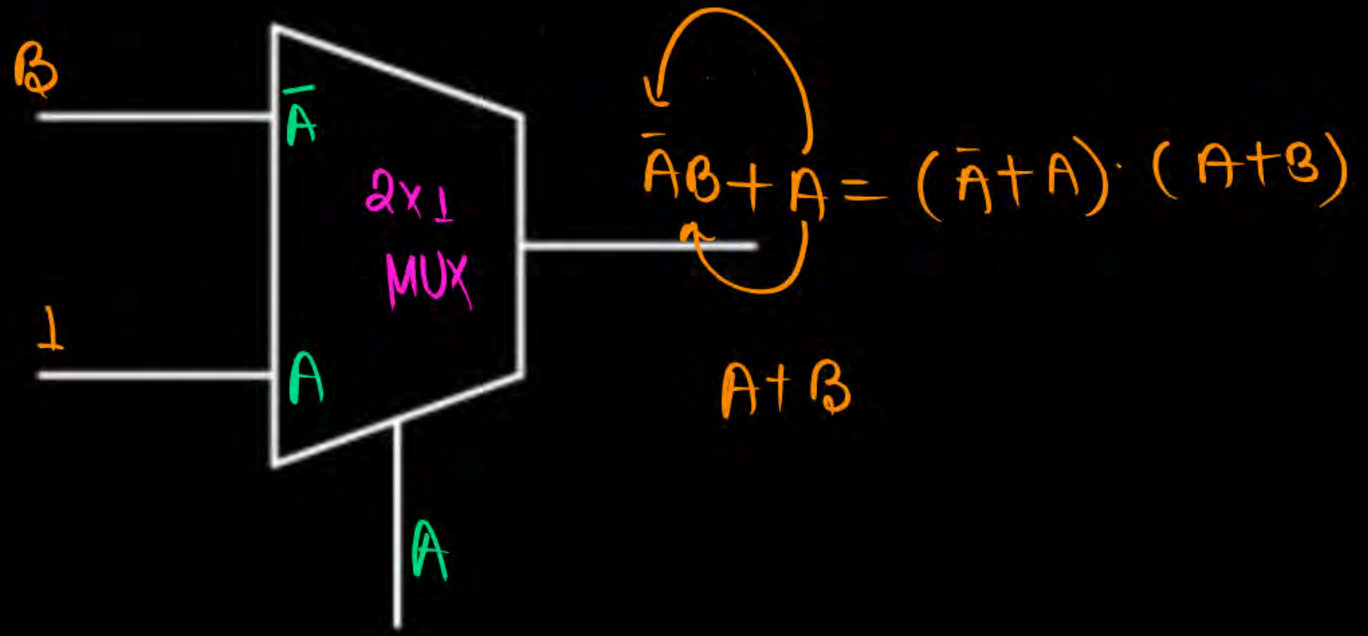




Type-2 MUX as a Universal Logic

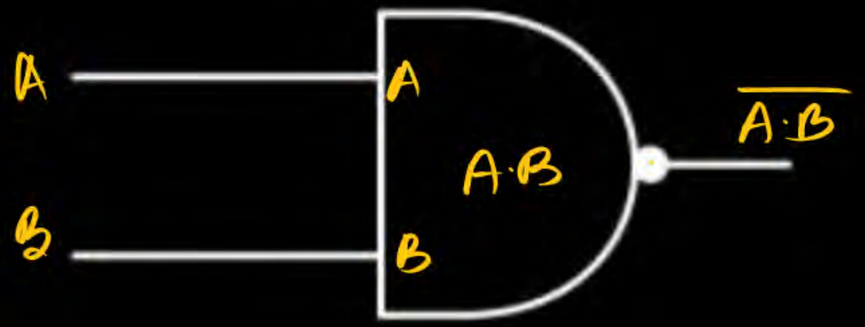
3. OR GATE

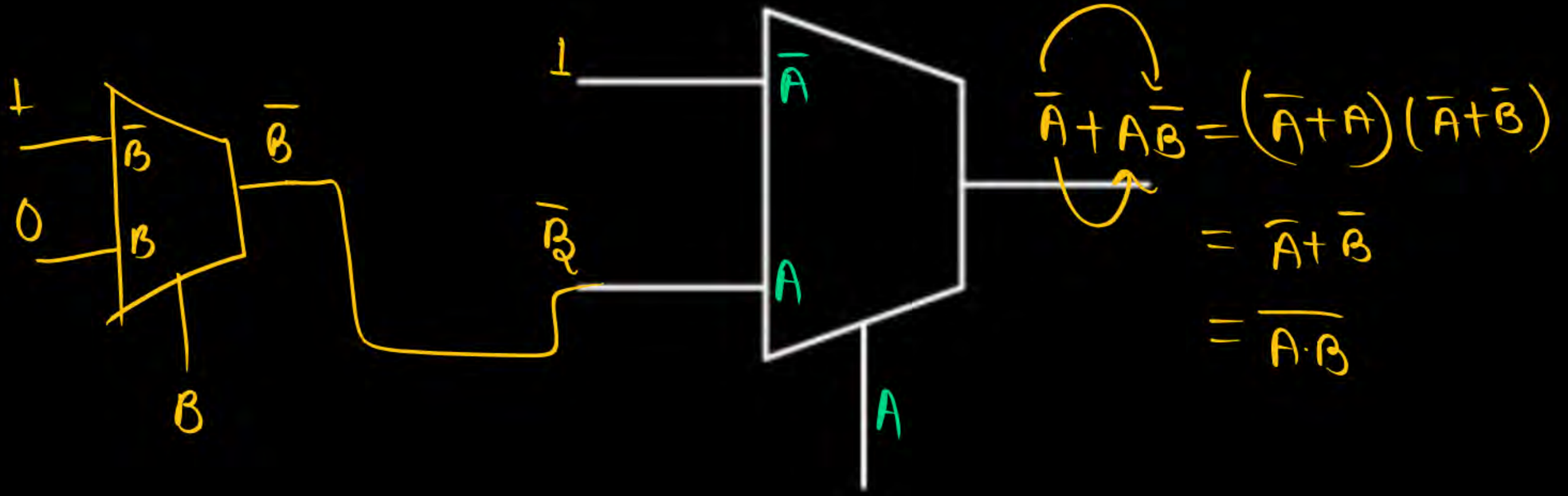




Type-2 MUX as a Universal Logic

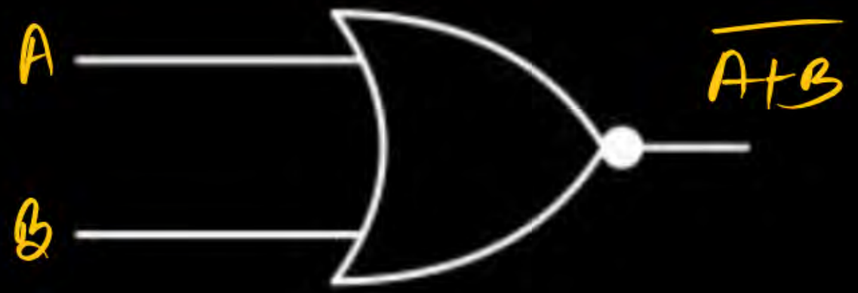
4. NAND GATE

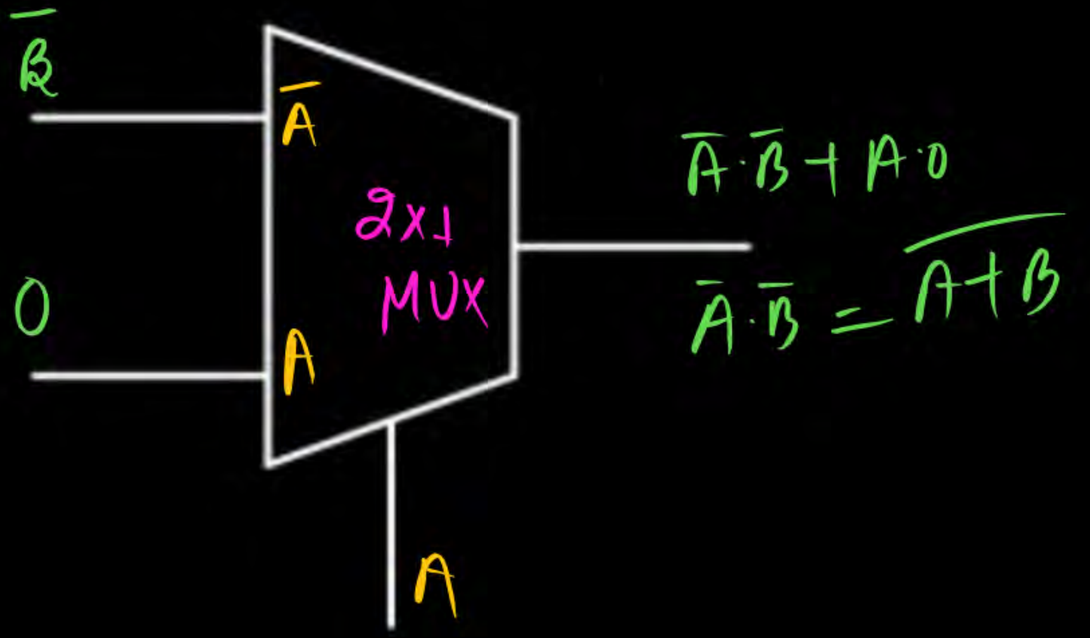




Type-2 MUX as a Universal Logic

5. NOR GATE

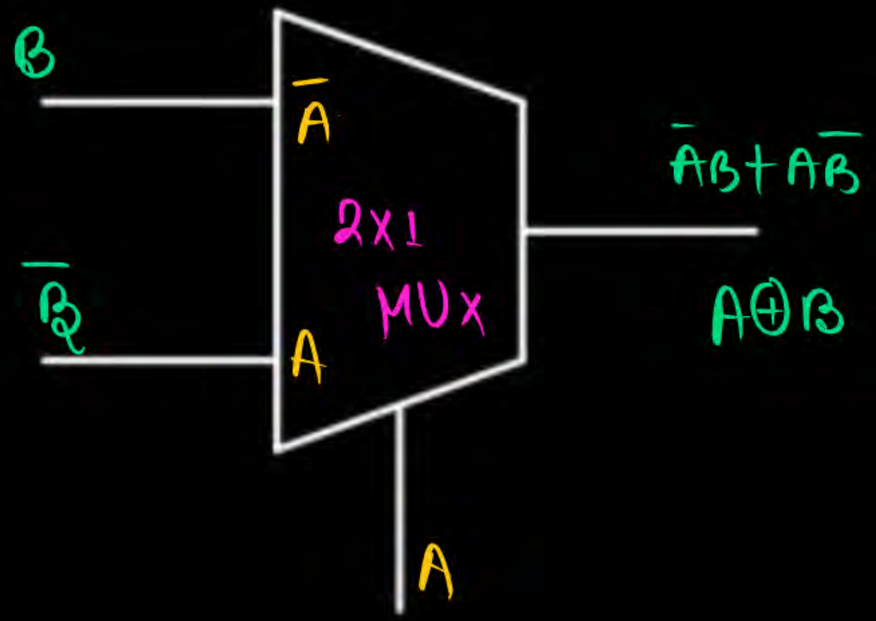




Type-2 MUX as a Universal Logic

6. X-OR GATE



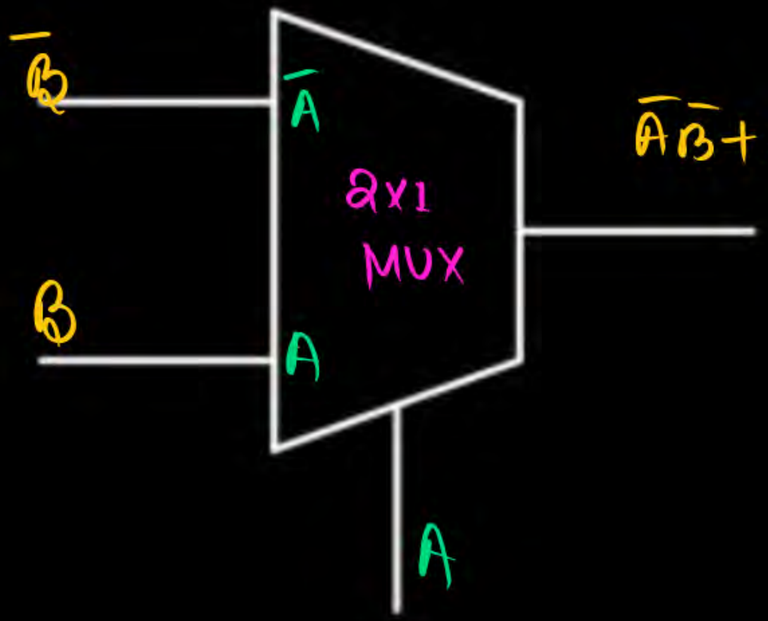


Type-2 MUX as a Universal Logic

7. X-NOR GATE



$A \odot B = \bar{A}\bar{B} + AB$



$\bar{A}\bar{B} + AB = A \odot B$

₹10 NOTE

↳ To Implement NOT, AND, OR **One** 2x1 MUX is required.

↳ To Implement NAND, NOR, XOR, X-NOR **Two** 2x1 MUX are required.

Type-3 Minimization

Q.

Find the output f.



ĀB
010→2
011→3

$= \Sigma m(0,2,3,6,7)$

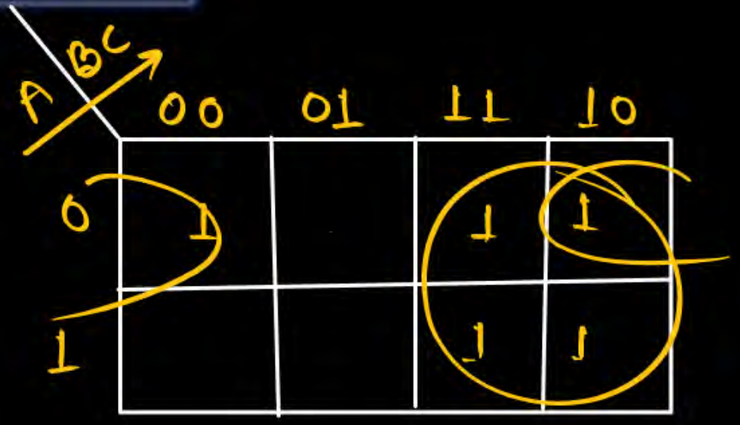
$f(A,B,C) = \bar{A}\bar{B}\bar{C} + \bar{A}B\cdot 1 + A\bar{B}\cdot 0 + AB\cdot 1$

$= \bar{A}\bar{B}\bar{C} + \bar{A}B + AB$

$= \bar{A}\bar{B}\bar{C} + \bar{A}B(\bar{C}+C) + AB(\bar{C}+C)$

$= \bar{A}\bar{B}\bar{C} + \bar{A}B\bar{C} + \bar{A}BC + AB\bar{C} + ABC$

$= \Sigma m(0,2,3,6,7)$



$= B + \bar{A}\bar{C}$ (Ans)

$\bar{A}\bar{C} = X$

$\bar{A}\bar{C}\bar{B} + B(\bar{A}+A)$

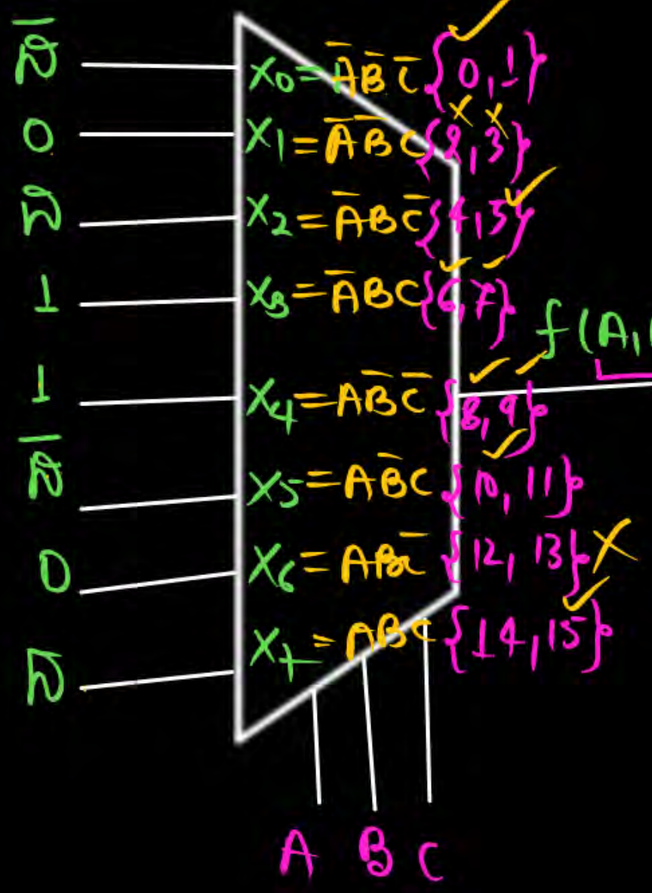
$B + \bar{A}\bar{C}\bar{B}$

$B + X\bar{B}$

$(B+X)(B+\bar{B})$

$B + X = B + \bar{A}\bar{C}$ Ans

Type-3 Minimization



$\bar{A}\bar{B}\bar{C}\bar{D}$
0 0 0 0

$\bar{A}\bar{B}\bar{C}D$

$\bar{A}BC$ से
0 1 1 0 → 6
0 1 1 1 → 7

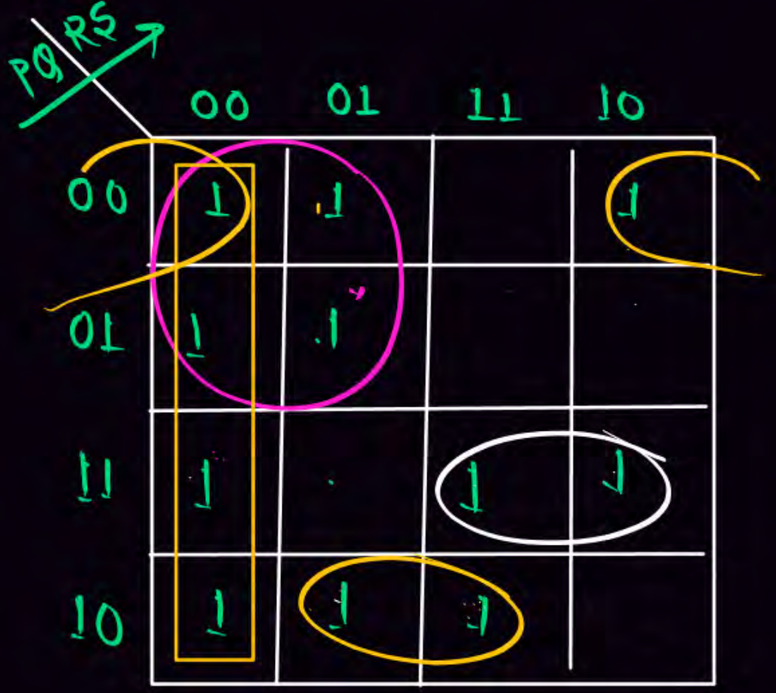
$A\bar{B}\bar{C}$ से
1 0 0 0 → 8
1 0 0 1 → 9

$f(A,B,C,D) = \sum m(0,5,6,7,8,9,10,15)$

AB \ CD	00	01	11	10
00	1			
01		1	1	1
11			1	
10	1	1		1

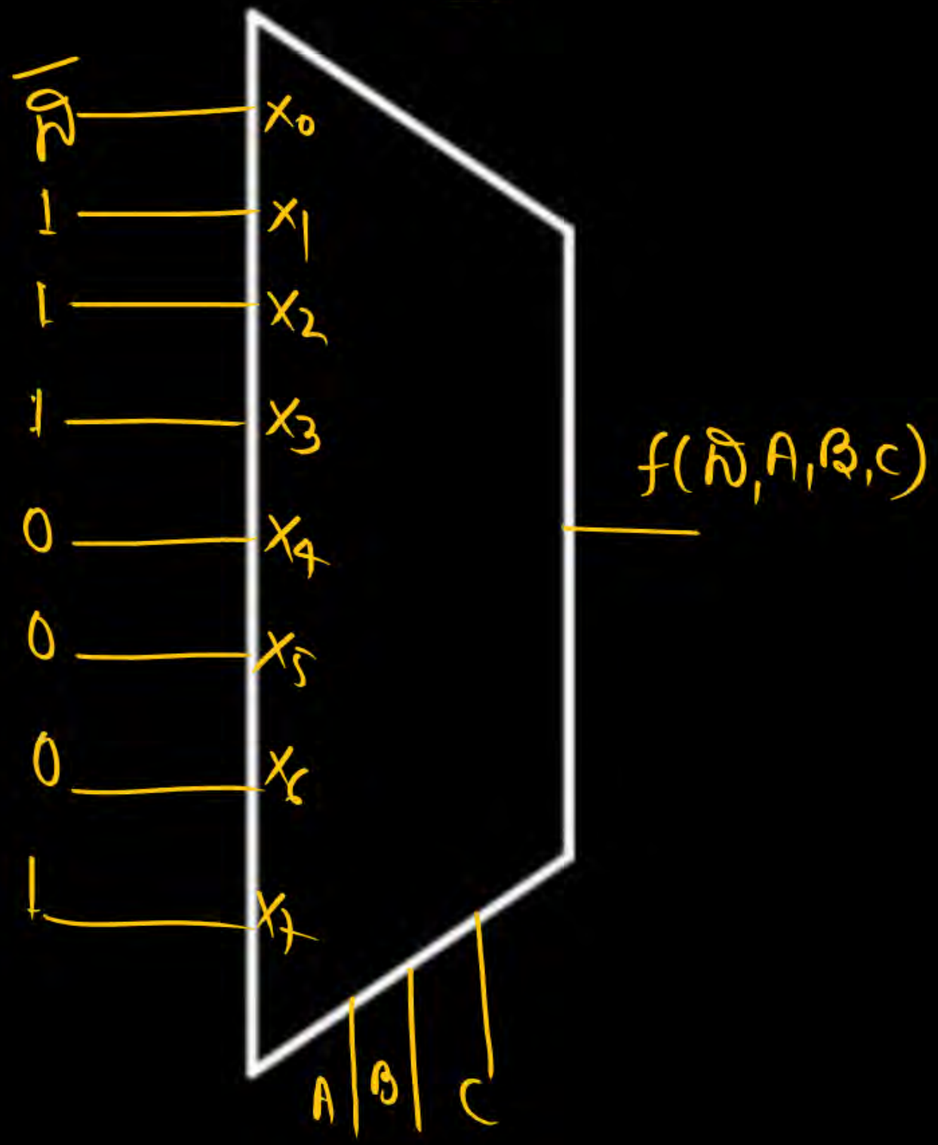


$$f(P,Q,R,S) = \Sigma m(0,1,2,4,5,8,9,11,12,14,15)$$



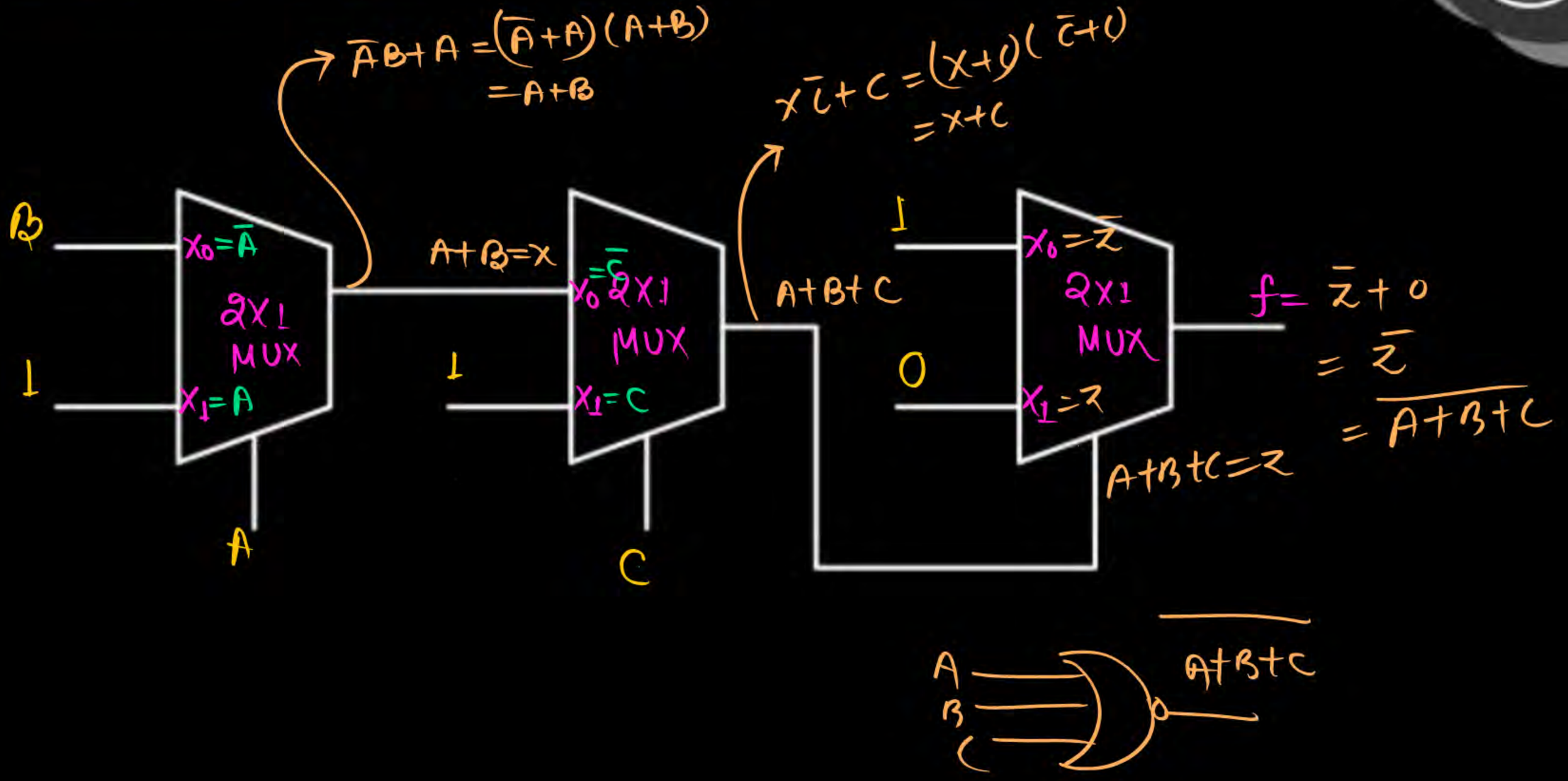
Type-3 Minimization

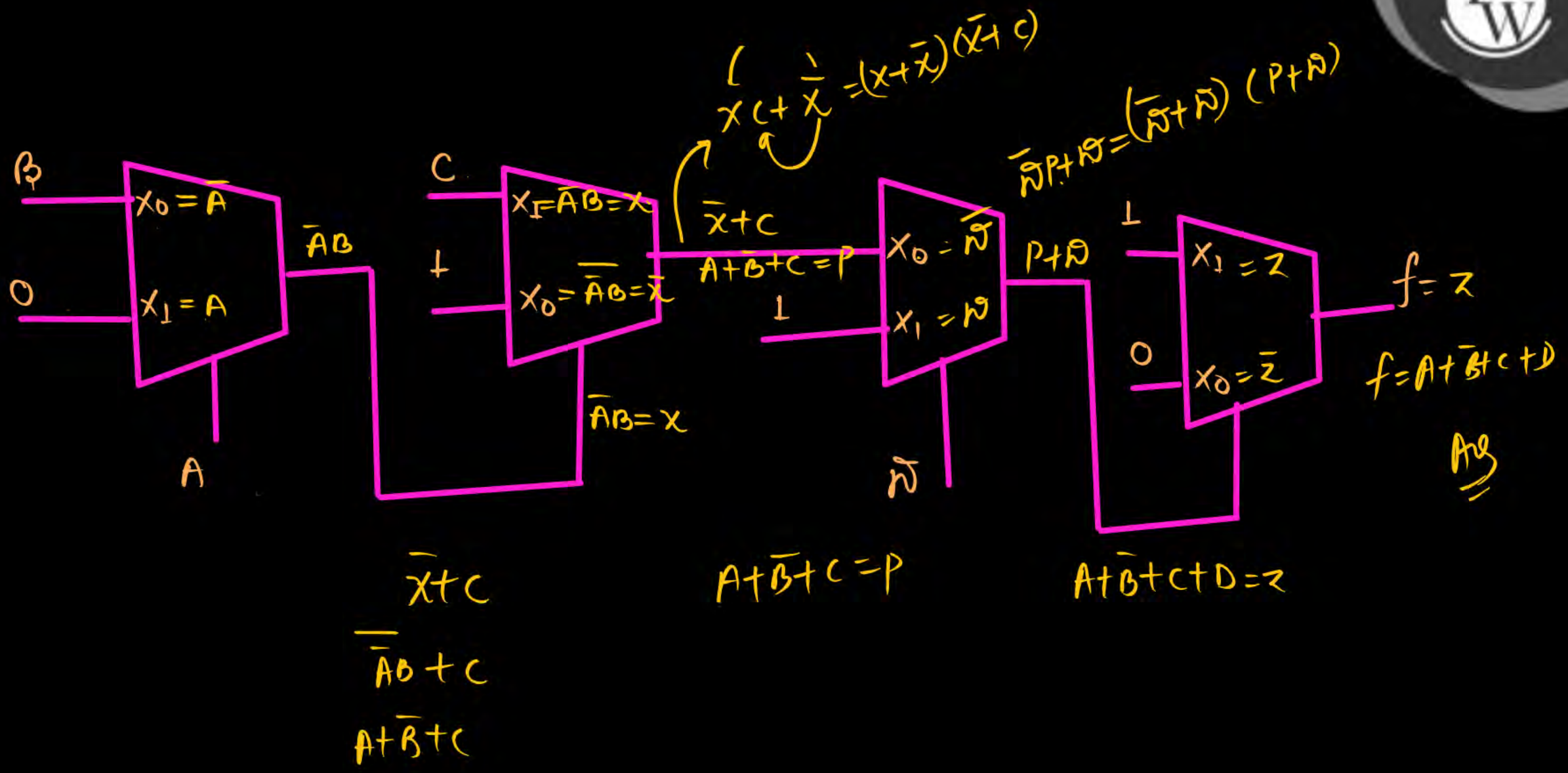
HW

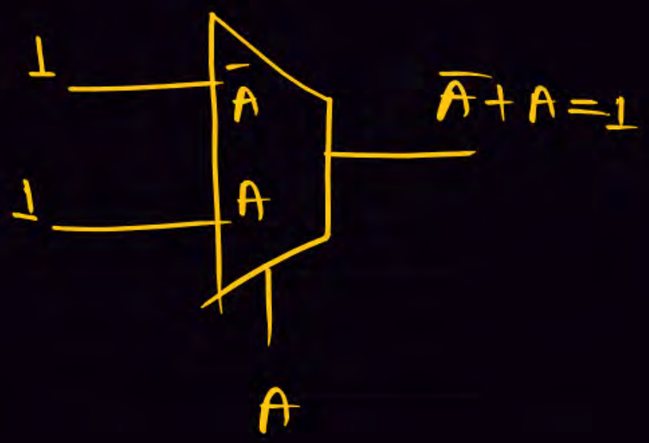


Type-4 Cascading of Mux

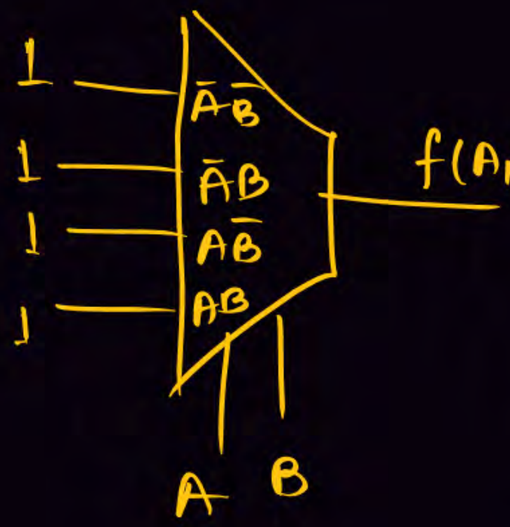
Ex




B
0
$X_0 = \bar{A}$
$X_1 = A$
A
$\bar{A}B$
C
1
$X_1 = \bar{A}B = x$
$X_0 = \bar{A}B = \bar{x}$
$\bar{A}B = x$
$\bar{x}+C$
$A+\bar{B}+C$
$\bar{A}B+C$
$x(\bar{x}+\bar{x}) = (x+\bar{x})(\bar{x}+C)$
$A+B+C = P$
1
$X_0 = \bar{N}$
$X_1 = N$
N
$A+\bar{B}+C = P$
$\bar{N}P+ND = (\bar{N}+N)(P+D)$
$P+D$
1
0
$X_1 = z$
$X_0 = \bar{z}$
$A+\bar{B}+C+D = z$
$f = z$
$f = A+\bar{B}+C+D$



$\bar{A} + A = 1$



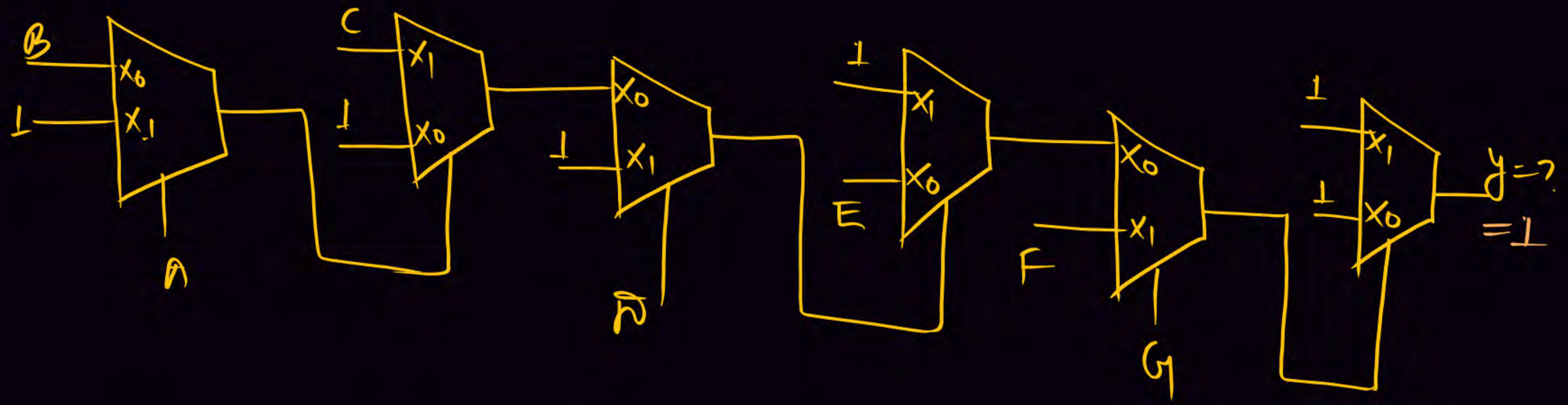
$f(A,B) = \bar{A}\bar{B} + \bar{A}B + A\bar{B} + AB$

$= \bar{A}(\bar{B}+B) + A(\bar{B}+B)$

$= \bar{A} + A$

$= 1$

Q =



TYPE-5 :→ Implementation of function :→

$f(A,B,C) = \bar{A}\bar{B}\bar{C} + \bar{A}B\bar{C} + AB\bar{C} + ABC = \Sigma m(0,2,6,7)$

Case(1) By 8X1 MUX.



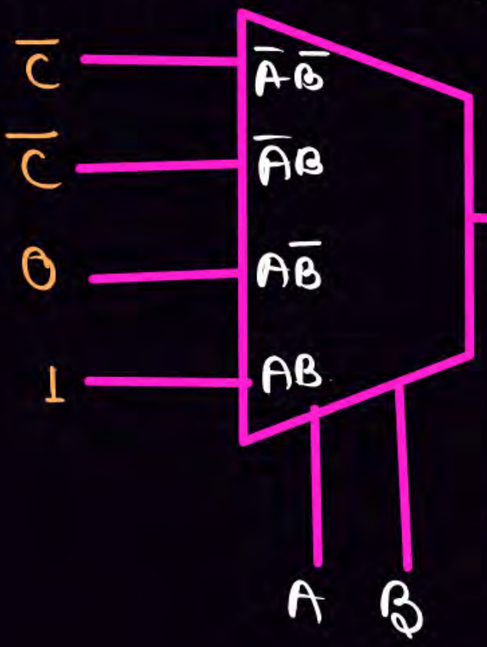
$f(A,B,C) = \bar{A}\bar{B}\bar{C} + \bar{A}B\bar{C} + AB\bar{C} + ABC$

$= \Sigma m(0,2,6,7)$

Case (2) By 4×1 MUX :→

$f(A,B,C) = \Sigma m(0,2,6,7)$

(1) AB as a select line



$f(A,B,C)$

$= \bar{A}\bar{B}\bar{C} + \bar{A}B\bar{C} + AB\cdot 1$

$= \bar{A}\bar{B}\bar{C} + \bar{A}B\bar{C} + AB(\bar{C}+C)$

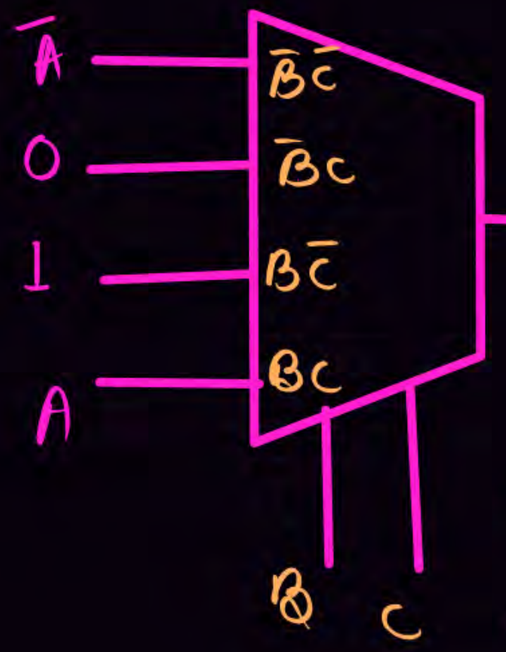
$= \bar{A}\bar{B}\bar{C} + \bar{A}B\bar{C} + AB\bar{C}$

$+ ABC$

	$\bar{A}\bar{B}$	$\bar{A}B$	$A\bar{B}$	AB
$\bar{C}$	$\bar{A}\bar{B}\bar{C}$ (0)	$\bar{A}B\bar{C}$ (2)	$A\bar{B}\bar{C}$ 4	$AB\bar{C}$ (6)
C	$\bar{A}\bar{B}C$ 1	$\bar{A}BC$ 3	$A\bar{B}C$ 5	ABC (7)
	$\bar{C}$	$\bar{C}$	0	1

(ii) BC as a select line.

$f(A,B,C) = \Sigma m(0,2,6,7)$



$f(A,B,C)$

$= \bar{A}B\bar{C} + B\bar{C} + ABC$

$= \bar{A}B\bar{C} + (\bar{A}+A)B\bar{C} + ABC$

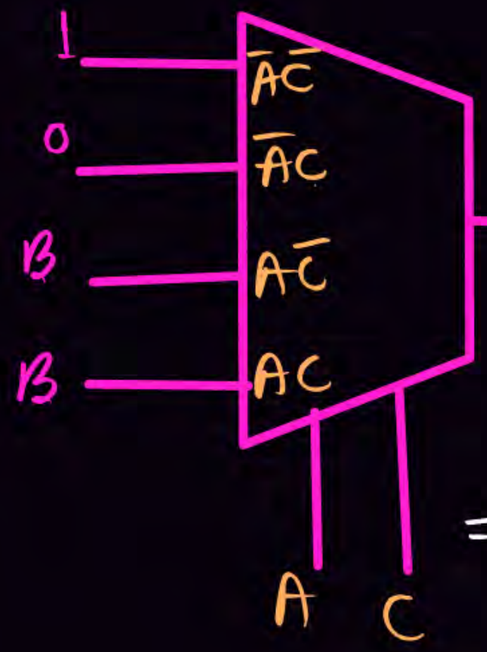
$= \bar{A}B\bar{C} + \bar{A}BC + AB\bar{C} + ABC$

$= \Sigma m(0,2,6,7)$

	$\bar{B}\bar{C}$	$\bar{B}C$	$B\bar{C}$	BC
$\bar{A}$	$\bar{A}\bar{B}\bar{C}$ (0)	$\bar{A}\bar{B}C$ 1	$\bar{A}B\bar{C}$ (2)	$\bar{A}BC$ 3
A	$A\bar{B}\bar{C}$ 4	$A\bar{B}C$ 5	$AB\bar{C}$ (6)	ABC (7)
	$\bar{A}$	0	1	A

(ii) AC as a select line

$f(A,B,C) = \Sigma m(0,2,6,7)$



$f(A,B,C) = \bar{A}\bar{C} + AB\bar{C} + ABC$

$\bar{A}(\bar{B}+B)\bar{C} + AB\bar{C} + ABC$

$= \bar{A}\bar{B}\bar{C} + \bar{A}B\bar{C} + AB\bar{C} + ABC$

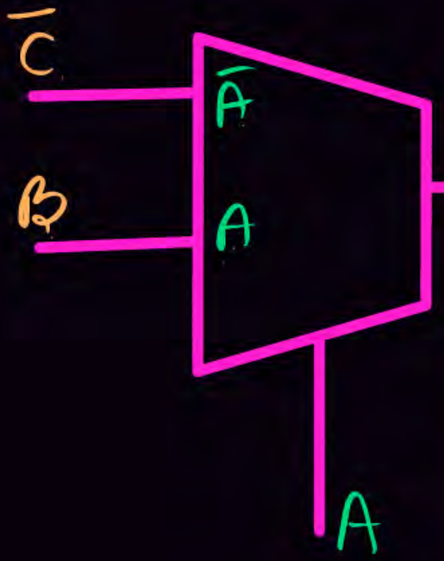
$= \Sigma m(0,2,6,7)$

	$\bar{A}\bar{C}$	$\bar{A}C$	$A\bar{C}$	AC
$\bar{B}$	$\bar{A}\bar{B}\bar{C}$ (0)	$\bar{A}\bar{B}C$ 1	$A\bar{B}\bar{C}$ 4	$A\bar{B}C$ 5
B	$\bar{A}B\bar{C}$ (2)	$\bar{A}BC$ 3	$AB\bar{C}$ (6)	ABC (7)
	1	0	B	B

case(3) By 2XL MUX :→

$f(A,B,C) = \Sigma m(0,2,6,7)$

(i) A as a select line



$f(A,B,C)$

$\bar{A}\bar{C} + AB$

$\bar{A}(\bar{B}+B)\bar{C} + AB(\bar{C}+C)$

$\bar{A}\bar{B}\bar{C} + \bar{A}B\bar{C} + AB\bar{C} + ABC$

$= \Sigma m(0,2,6,7)$

input combination	$\bar{A}$	A
$\bar{B}\bar{C}$	$\bar{A}\bar{B}\bar{C}$ (0)	$A\bar{B}\bar{C}$ 4
$\bar{B}C$	$\bar{A}\bar{B}C$ 1	$A\bar{B}C$ 5
$B\bar{C}$	$\bar{A}B\bar{C}$ (2)	$AB\bar{C}$ (6)
BC	$\bar{A}BC$ 3	ABC (7)
	$B\bar{C} + B\bar{C}$	$B\bar{C} + BC$
	$\bar{C}(\bar{B}+B)$	$B(\bar{C}+C) = B$
	$= \bar{C}$	

(ii) B as a select line.

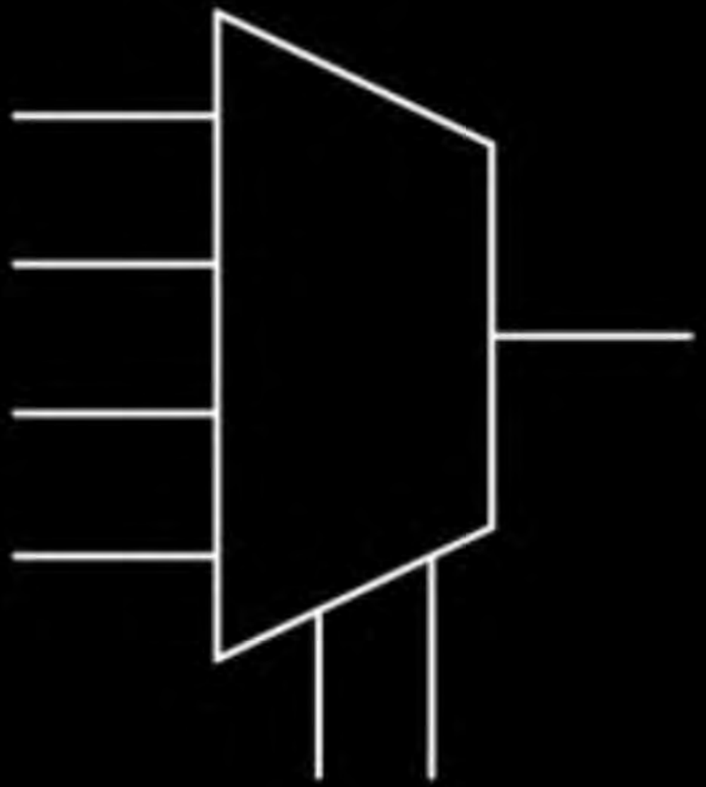
(iii) C as a select line.

By using 4 × 1 MUX

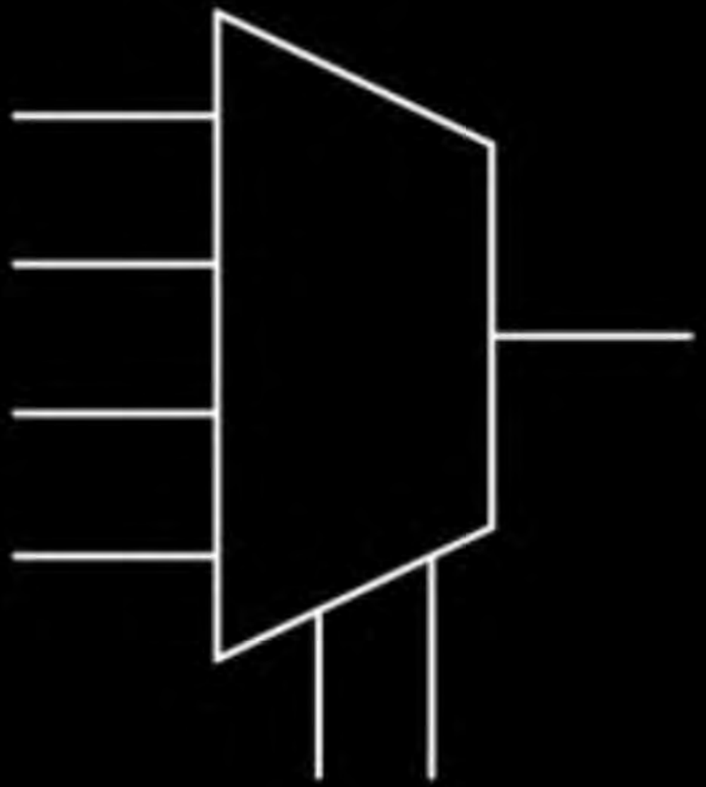
$f(A, B, C) = \bar{A}\bar{B}\bar{C} + \bar{A}\bar{B}C + A\bar{B}\bar{C} + ABC$

1. AB as a select line
2. BC as a select line
3. AC as a select line

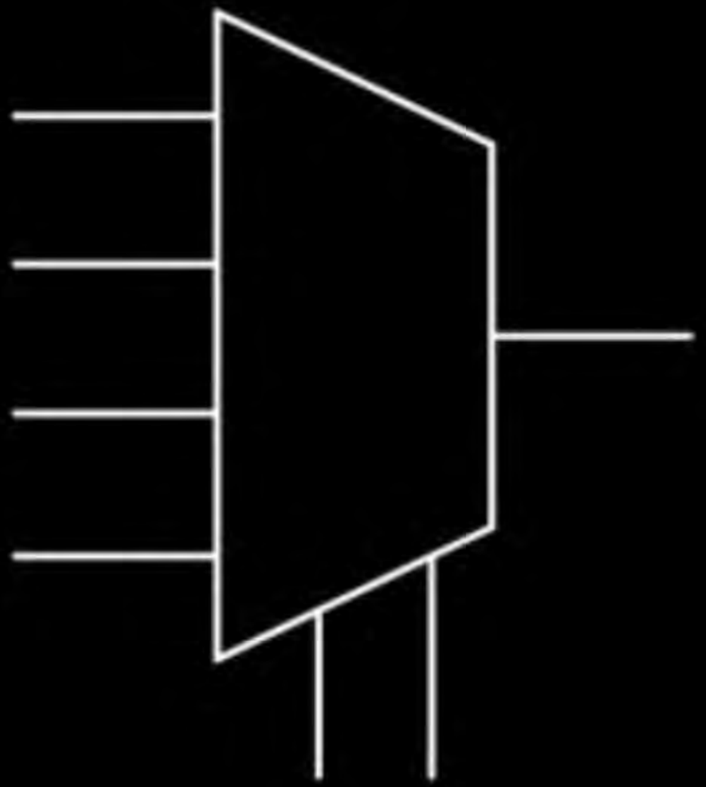
1. AB is select line $f(A, B, C) = \bar{A}\bar{B}\bar{C} + \bar{A}\bar{B}C + A\bar{B}\bar{C} + ABC$



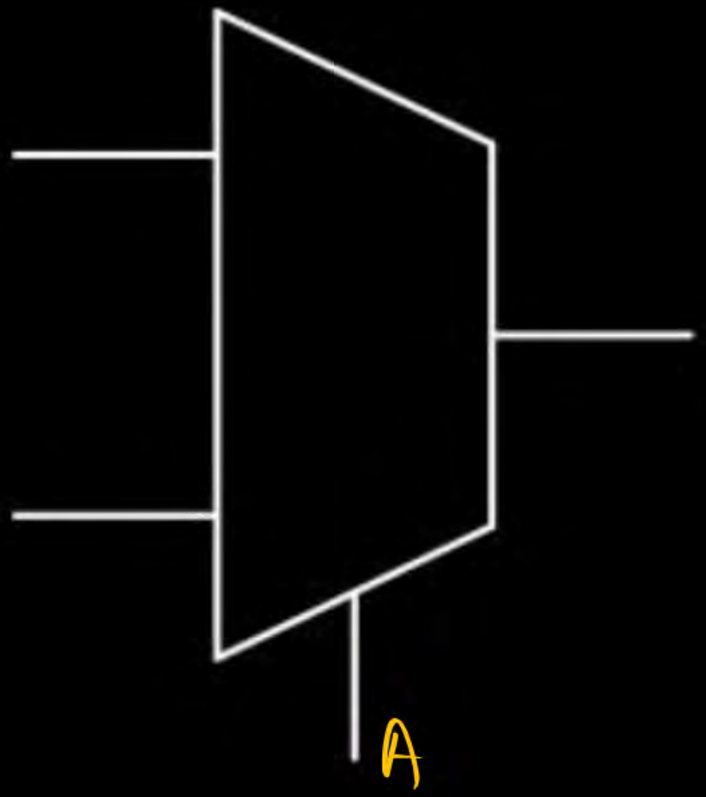
2. BC is select line $f(A, B, C) = \bar{A}\bar{B}\bar{C} + \bar{A}\bar{B}C + A\bar{B}\bar{C} + ABC$



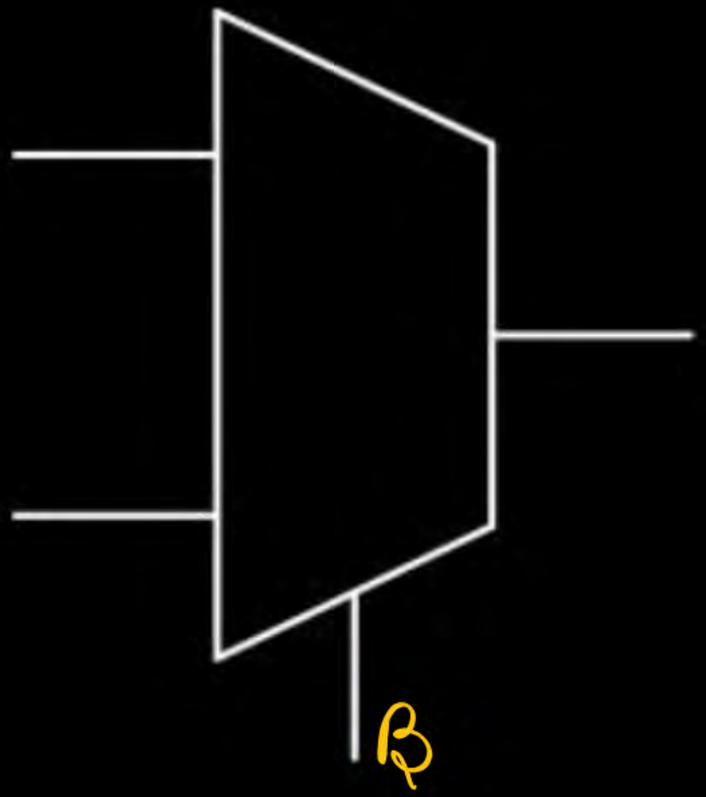
3. AC is select line $f(A, B, C) = \bar{A}\bar{B}\bar{C} + \bar{A}\bar{B}C + A\bar{B}\bar{C} + ABC$



By using 2 × 1 MUX



By using 2 × 1 MUX



Q.

$f(A, B, C, D) = \sum m(0, 1, 3, 5, 7, 9, 12, 15)$

(i) 8×1 MUX

1. ABD as a select line
2. ACD as a select line

(ii) 4×1 MUX

(i) AB as a select line

(ii) AD as a select line

(iii) AC as a select line

Thank you

GW

Soldiers !

